

Midterm 2

Marcus Owens

1. Unit 4:

1) B: 45 rpm, 4.7 radians sec^{-1}

2) 2.2 ms^{-2}

3) D: $\omega \rightarrow 10\omega$

4) B: 0.53 N

5) C: 42 ms^{-1}

2. Unit 5:

1) D: 200 Nm, 37 degrees

2) a) $W = F \cdot d$

$$W = 1000 \text{ N} \cdot 0.1 \text{ m}$$

$$\boxed{W = 100 \text{ J}}$$

b) $\boxed{\Delta K = 100 \text{ J}}$

c) $\Delta K = \frac{1}{2} m v_f^2 \rightarrow v_f = \sqrt{\frac{2\Delta K}{m}}$

$$\boxed{v_f = 36.5 \text{ m/s}}$$

d) $R = \frac{v_o^2 \sin(2\theta)}{g}$

$$R = \frac{(36.5)^2}{9.81}$$

$$\boxed{R = 135.9 \text{ m}}$$

3.) a. $F_k = \mu_k \cdot N$

$N = 40 \times 9.81 = 392.4 \text{ N}$

$W = F_k \cdot d$

$W = 19.62 \times 40$

$F_n = .05 \times 392.4 = 19.62 \text{ N}$

$W = 784.8 \text{ J}$

b. $p = \frac{W}{t}$
 $= \frac{784.8}{30 \text{ s}}$

$p = 26.16 \text{ J/s}$

4) 0, \$2.33

Unit 6:

1) 6.0 m/s

2) $VF = \frac{p}{m_1 + m_2} = \frac{p}{m_1 + m_2}$
 $VF = 0 \text{ m/s}$

3) a. impulse $= F \cdot \Delta t$
 $= 4000 \times 0.2 \text{ s}$
 $= 800 \text{ N} \cdot \text{s}$

b. $VF = \frac{800}{200} + 2.00$
 $VF = 6.00 \text{ m/s}$

4) c. elastic

5.) a. $(30,000)(0.85) + (110,000)(0) = (30,000 + 110,000)V_f$

$$V_f = \frac{25500}{140000}$$

$$\boxed{V_f = 0.182 \text{ m/s}}$$

b. $KE = \frac{1}{2}mv^2$

$$KE = \frac{1}{2}(30000)(0.85)^2$$

$$KE = 10837.5 \text{ J} \rightarrow \text{before}$$

$$KE = \frac{1}{2}(140000)(0.182)^2$$

$$KE = 2318.68 \rightarrow \text{after}$$

$$10837.5 - 2318.68 = \boxed{8518.82 \text{ J}}$$

Unit 7

1.) A: remain motionless

$$2) W = m \cdot g = 20 \times 9.81 = 196.2 \text{ N} = F_1 + F_2$$

$$F_1 = \frac{98.1}{2}$$

$$196.2 - 49.05 =$$

$$\boxed{F_1 = 49.05 \text{ N}}$$

$$\boxed{F_2 = 147.15 \text{ N}}$$

Unit 8:

1) a. $I = \frac{1}{2}MR^2$

$$I = \frac{1}{2}(1)(0.01)^2$$

$$\boxed{I = 0.0032 \text{ kg} \cdot \text{m}^2}$$

2) a. $-2.4 \hat{k} \text{ N} \cdot \text{m}$

b. $-12 \hat{j} \text{ N}$

b. $I = I_a$

~~$$I = \frac{1}{2}MR^2$$~~

$$T = (0.0032)(2)$$

$$\boxed{T = 0.0064 \text{ N} \cdot \text{m}}$$